

CS601 Midterm Preparation Lec 1-113 MCQ'S

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1-What is the primary definition of telecommunications?

- a) Communication using signals
- b) Communication over short distances
- c) Communication at a distance**
- d) Communication through written messages

2-What is involved in data communications?

- A) Exchange of data between two devices via transmission media**
- B) Exchange of ideas through speech
- C) Exchange of data between multiple people
- D) Exchange of messages through phone calls

3-Which of the following is NOT one of the key components for the effectiveness of a data communication system?

- a) Delivery
- b) Accuracy
- c) Simplicity**
- d) Jitter

4-The exchange of data between two devices is referred to as:

- a) Telecommunications
- b) Data communications**
- c) Local communications
- d) Distance communications

5-Jitter is a factor in determining the effectiveness of which system?

- a) Television system
- b) Data communication system**
- c) Telephone system
- d) Telegraph system

6- Which of the following describes a simplex communication mode?

- a) Data flows in both directions simultaneously
- b) Data flows in both directions, but one at a time
- c) Data flows in one direction only**
- d) Data does not flow at all

7-What does a full-duplex communication system allow?

- a) Data flows in one direction only
- b) Data flows in both directions simultaneously**
- c) Data flows in both directions, but one at a time
- d) Data flows only when initiated by a server

8-What is a network?

- a) A device that sends and receives data
- b) An interconnection of a set of devices capable of communication**
- c) A system used for half-duplex communication
- d) A security system for data communication

9-Which of the following is NOT a factor in network performance?

- a) Throughput
- b) Delay
- c) Security**
- d) Reliability

10-The physical topology of a network refers to:

- a) The physical layout of links and nodes**
- b) The communication protocol
- c) The type of data being transmitted
- d) The method of securing data communication

11-Which of the following is a type of physical topology?

- a) Ring
- b) Star
- c) Bus
- d) All of the above**

12-Local Area Networks (LANs) typically connect hosts within:

- a) A city
- b) A single office, building, or campus**
- c) Multiple countries
- d) A wide geographical span

13-Which of the following is true about Wide Area Networks (WANs)?

- a) They connect devices within a single building

b) They cover a larger geographical area than LANs

- c) They are always privately owned
- d) They only connect two devices

14-Local Area Networks (LANs) are typically:

a) Privately owned

- b) Run by telecommunication companies
- c) Used to cover a town or city
- d) Used to connect devices globally

15-Which of the following networks spans across countries or even globally?

a) LAN

b) WAN

- c) Host network
- d) Home network

16-Which devices are typically interconnected in a WAN?

a) PCs and printers

b) Switches, routers, or modems

- c) Only computers
- d) None of the above

17-Network classification can be based on which of the following factors?

a) Size, geographical coverage, and ownership

- b) Protocol
- c) Number of hosts
- d) Type of communication

18- What is the correct definition of an internet (with lowercase "i")?

a) Two or more networks that can communicate with each other

- b) The global network connecting billions of devices
- c) A communication protocol
- d) A local network within a building

19-The Internet (with uppercase "I") refers to:

a) Two connected networks

b) Thousands of interconnected networks

- c) A private network for businesses
- d) A single, isolated network

20-What type of communication did telegraph and telephone networks use before 1960?

- a) Packet-switched communication
- b) Variable-rate communication

- c) Digital communication
- d) Constant-rate communication**

21-The ARPANET is significant because it was the first network to use:

- a) Constant-rate communication
- b) Packet-switched communication**
- c) Telephone lines for data transfer
- d) Radio waves

22-What is a protocol in networking?

- a) A network device
- b) A type of network
- c) Rules that both the sender and receiver must follow to communicate effectively**
- d) A method of increasing network speed

23-Protocol Layering is necessary for:

- a) Simple communication
- b) Complex communication**
- c) Constant-rate communication
- d) Local communication only

24-Which of the following is an advantage of protocol layering?

- a) Modularity**
- b) Increased communication delays
- c) High complexity
- d) Increased cost

26- Which of the following describes one of the two principles in networking?

- a) One-directional communication
- b) Bidirectional communication, where each layer performs opposite tasks in each direction**
- c) Two layers performing the same task
- d) Layers communicating randomly

27- The TCP/IP Protocol Suite was officially chosen as the protocol for the Internet in which year?

- a) 1973
- b) 1977
- **c) 1983**
- d) 1990

28-What is true about each layer in the TCP/IP Protocol Suite?

- a) Each layer is identical across all protocols
- **b) Each layer provides specific functionality**
- c) Each layer changes dynamically
- d) Each layer is responsible for security

29- In the TCP/IP Protocol Suite, what are the two essential addresses needed for communication?

- **a) Source Address and Destination Address**
- b) Physical Address and Logical Address
- c) MAC Address and IP Address
- d) Router Address and Subnet Mask

30- Which layer in the TCP/IP Protocol Suite is an exception to addressing by layer?

- **a) Data Link Layer**
- b) Network Layer
- c) Physical Layer
- d) Transport Layer

31-The OSI Model was introduced by the ISO in the:

- a) Early 1970s
- **b) Late 1970s**
- c) 1980s
- d) 1940s

32- The OSI Model consists of how many layers?

- a) 5 layers
- b) 6 layers
- **c) 7 layers**
- d) 4 layers

33-The International Organization for Standardization (ISO) was established in:

- a) 1977
- **b) 1947**
- c) 1983
- d) 1950

34- Which layer of the OSI model is responsible for the physical transmission of bits?

A. Physical

B. Data Link

C. Network

D. Transport

35-What layer handles the establishment, management, and termination of sessions between communicating applications?

A. Network

B. Transport

C. Session

D. Presentation

36-Which layer is concerned with the syntax, semantics, and encoding of data?

A. Application

B. Presentation

C. Session

D. Transport

37-The layer responsible for routing packets from source to destination is:

A. Data Link

B. Network

C. Transport

D. Session

38-Which layer provides reliable end-to-end delivery of data?

A. Network

B. Transport

C. Session

D. Presentation

39-The highest layer in the OSI model that directly interacts with the user is:

A. Network

B. Transport

C. Presentation

D. Application

40-Error detection and correction occur primarily at which layer?

A. Physical

B. Data Link

C. Network

D. Transport

41-Which layer is responsible for framing data into packets?

A. Physical

B. Data Link

C. Network

D. Transport

42-The OSI model is a reference model for:

A. Hardware

B. Software

C. Network protocols

D. Applications

43-Which layer is responsible for translating data formats between different systems?

A. Network

B. Transport

C. Presentation

D. Application

44- Which layer in the OSI model does not have a direct equivalent in the TCP/IP protocol suite?

A. Application

B. Presentation

C. Session

D. Transport

45-A signal that repeats itself after a fixed interval of time is called:

A. Non-periodic

B. Periodic

C. Analog

D. Digital

46-A signal that does not repeat itself after any interval of time is called:

A. Non-periodic

B. Periodic

C. Analog

D. Digital

47-Which type of signal can be both periodic and non-periodic?

A. Analog

B. Digital

C. Periodic only

D. Non-periodic only

48-The time taken for a periodic signal to complete one full repetition is called:

A. Frequency

B. Period

C. Amplitude

D. Wavelength

49-A periodic signal has a pattern, while a non-periodic signal does not.

A. pattern ... does not

B. does not ... has a pattern

C. continuous ... discrete

D. discrete ... continuous

50-Which of the following is an example of a periodic signal?

A. Human speech

B. A sine wave

C. A single pulse

D. White noise

51-Which of the following is an example of a non-periodic signal?

A. Music

B. A square wave

C. A triangular wave

D. A constant DC voltage

52-Digital signals are discrete representations of analog signals.

A. discrete

B. continuous

C. analog

D. periodic

53-The frequency of a periodic signal is the reciprocal of its period.

A. reciprocal

B. square root

C. logarithm

D. exponential

54-A periodic signal can be represented by a Fourier series.

A. periodic

B. non-periodic

C. analog

D. digital

55-Which two layers from the OSI Model are missing in the TCP/IP model?

a) Transport and Session

b) Presentation and Session

c) Network and Physical

d) Data Link and Network

56-In the TCP/IP model, the Application Layer is equivalent to which layers in the OSI Model?

a) Application, Presentation, and Session

b) Transport and Network

c) Data Link and Physical

d) Session and Data Link

57-One reason OSI did not replace TCP/IP is:

a) OSI was completed before TCP/IP

b) OSI was completed when TCP/IP was already fully in place

c) OSI had better performance than TCP/IP

d) OSI had fewer layers than TCP/IP

58-Another reason OSI did not replace TCP/IP is that:

a) TCP/IP lacked important features

b) Some layers in OSI were not fully defined

c) OSI was easier to implement

d) OSI had better-defined protocols

59-Why was the performance of TCP/IP preferred over OSI?

a) TCP/IP had better performance than OSI

b) OSI had more defined standards

c) OSI was faster in implementation

d) Both had similar performance

60- The frequency of a sine wave is the reciprocal of its period.

A. frequency

B. amplitude

C. phase

D. wavelength

61-The unit of frequency is:

A. Hertz (Hz)

B. Seconds (s)

C. Meters (m)

D. Watts (W)

62-A sine wave with a period of 1 millisecond has a frequency of:

A. 1 kHz

B. 1 MHz

C. 1 GHz

D. 1 Hz

63-A sine wave with a frequency of 100 MHz has a period of:

A. 100 ns

B. 10 ns

C. 1 μ s

D. 10 μ s

64-The relationship between period and frequency is:

A. Frequency = Period / 2

B. Frequency = Period * 2

C. Frequency = 1 / Period

D. Frequency = Period

65-The higher the frequency of a sine wave, the shorter its period.

A. shorter

B. longer

C. higher

D. lower

66-A sine wave with a frequency of 1 GHz has a period of:

A. 1 μ s

B. 10 ns

C. 1 ns

D. 10 μ s

- A sine wave with a frequency of 1 kHz has a period of:

A. 1 ms

B. 10 ns

C. 1 ns

D. 10 μ s

- A sine wave with a frequency of 1 THz has a period of:

A. 1 ms

B. 10 ps

C. 1 ps

D. 10 μ s

- A sine wave with a frequency of 1 mHz has a period of:

- A. 1 ms
- B. 10 ns
- C. 10 us

D. 1 μ s

67-The amplitude of a sine wave is related to its maximum value.

- A. frequency
- B. period
- C. phase

D. amplitude

68-A sine wave with a frequency of 1 kHz has a period of:

A. 1 ms

- B. 10 ms
- C. 100 ms
- D. 1 s

69-The phase of a sine wave is related to its starting point.

A. phase

B. frequency

C. period

D. amplitude

70- Bandwidth is a measure of:

A. The speed of a network

B. The distance between two points

C. The range of frequencies in a signal

D. The amount of data stored on a device

71-Bandwidth can be measured in:

A. Hertz (Hz)

B. Bits per second (bps)

C. Both A and B

D. Neither A nor B

72-A wider bandwidth means:

A. A higher frequency range

B. A lower frequency range

C. A smaller range of frequencies

D. No change in the frequency range

73- What is a simple periodic analog signal?

a) A sine wave

- b) A composite wave
- c) A square wave
- d) A signal composed of multiple waves

74-A composite periodic analog signal is made up of:

- a) A single sine wave

b) Multiple sine waves

- c) Square waves only
- d) A constant signal with no variation

75-The period (T) of a sine wave represents:

a) The amount of time required to complete one cycle

- b) The number of cycles in one second
- c) The amplitude of the wave
- d) The phase shift of the wave

76-If the frequency of a signal is 60 Hz, what is its period?

- a) 60 seconds

b) 1/60 seconds or approximately 16.67 milliseconds

- c) 60 milliseconds
- d) 100 milliseconds

77-What does the phase or phase shift of a wave indicate?

- a) The frequency of the wave

b) The amplitude of the wave

c) The position of the waveform relative to time 0

d) The power of the wave

78-Which of the following is an important characteristic that measures network performance?

a) Bandwidth

b) Frequency

c) Period

d) Amplitude

79-Bandwidth can be measured in two different ways. Which of the following are correct?

a) Bandwidth in Hertz and bandwidth in bits per second

b) Bandwidth in meters and seconds

c) Bandwidth in frequency and time

d) Bandwidth in waves and cycles

80- The bandwidth of a composite signal is the difference between:

a) Amplitude and phase

b) The highest and lowest frequencies

c) Period and frequency

d) Voltage levels

81-In a digital signal, how can a 1 and a 0 be represented?

a) 1 as a positive voltage and 0 as zero voltage

- b) 1 as a high frequency and 0 as a low frequency
- c) 1 as a sine wave and 0 as a cosine wave
- d) 1 as a positive frequency and 0 as a negative frequency

82-What is true about digital signals with more than two levels?

a) They are used to reduce noise

b) They can send more than one bit for each level

- c) They are always periodic
- d) They are only used in analog transmissions

83-Bit rate is defined as:

a) The number of frequencies sent in one second

b) The number of bits sent in one second

- c) The difference between the highest and lowest bit values
- d) The amount of data stored in memory

84-Most digital signals are non-periodic, which means:

a) Period and frequency are not appropriate characteristics

- b) They are composed of sine waves
- c) They have a constant frequency
- d) They require modulation

85-Why is modulation used in communication systems?

a) To increase the amplitude of the signal

c) To compress data

b) To change a digital signal to an analog signal for transmission

d) To reduce signal impairments

86-What type of channel does modulation allow us to use?

a) A zero-bandwidth channel

b) A bandpass channel, where the bandwidth does not start from zero

c) A high-frequency channel

d) A low-frequency channel

87-One common problem with transmission media is that they cause:

a) Signal strengthening

b) Signal impairments

c) Increased bandwidth

d) Faster bit rate

88-Bandwidth can be expressed in which of the following units?

a) Hertz and bits per second

b) Amperes and volts

c) Bytes and kilobytes

d) Cycles and time

89-The changing of a digital signal to an analog signal for transmission is primarily done through:

a) Amplification

b) Encoding

c) Modulation

d) Decoding

90- What does distortion refer to in signal processing?

a) Changes in the form or shape of the signal

b) An increase in signal strength

c) A reduction in signal quality

d) The loss of a signal entirely

91-Distortion can occur in:

a) Only analog signals

b) Only digital signals

c) A composite signal made of different frequencies

d) Constant signals

92-Each signal component in a composite signal has its own:

a) Amplitude

b) Propagation speed

c) Frequency

d) Voltage level

93-What may cause a difference in phase in signal transmission?

a) Equal propagation speeds

b) Differences in delay not matching the period duration

c) Similar frequencies

d) Constant amplitude

94-Which of the following is NOT a type of noise that can corrupt a signal?

a) Thermal noise

b) Induced noise

c) Crosstalk

d) Distortion noise

95-Thermal noise is caused by:

a) External electromagnetic fields

b) Random motion of electrons in a wire

c) Nearby motors

d) Signal reflection

96-What is crosstalk?

a) The effect of environmental interference

b) The effect of one wire on another wire

c) A type of thermal noise

d) The loss of a signal over distance

97-Signal to Noise Ratio (SNR) is used to:

a) Measure the voltage level of a signal

b) Calculate the distance of signal transmission

c) Find the theoretical bit rate limit of a signal

d) Determine the phase of a signal

98-The bandwidth of a subscriber line for voice or data is:

a) 2 kHz

b) 4 kHz

c) 6 kHz

d) 8 kHz

99-If the bandwidth of a line is increased to 8 kHz, the possible data transmission rate can reach:

a) 56,000 bps

b) 112,000 bps

c) 64,000 bps

d) 128,000 bps

100- Throughput is defined as:

a) The total bandwidth of a network

b) How fast we can actually send data through a network

c) The maximum data that can be stored

d) The total number of connections in a network

101-Bandwidth is different from throughput because:

a) They are both the same

b) Bandwidth is always greater than throughput

c) A link may have a bandwidth of B bps, but we can only send T bps, with T always less than B

d) Throughput is not measurable

102-Latency refers to:

a) The amount of data that can be sent

b) How long it takes for an entire message to arrive at the destination

c) The speed of the data transfer

d) The distance between two points

103-The total latency is composed of which of the following components?

a) Only transmission time

b) Only queuing time

c) Propagation time, transmission time, queuing time, and processing delay

d) Only processing delay

104-What is the formula for calculating latency?

a) Latency = propagation time + transmission time + queuing time + processing delay

b) Latency = bandwidth / throughput

c) Latency = distance / speed

d) Latency = transmission time - propagation time

105-If the distance between two points is 12,000 km and the propagation speed is 2.4×10^8 m/s, what is the propagation time?

a) 30 ms

b) 50 ms

c) 100 ms

d) 20 ms

106-For a 2.5-KB message with a bandwidth of 1 Gbps, the transmission time can be calculated as:

a) 1 ms

b) 0.020 ms

c) 0.5 ms

d) 5 ms

107-What is the propagation time for a distance of 12,000 km given the speed of light is 2.4×10^8 m/s?

a) 20 ms

b) 100 ms

c) 50 ms

d) 70 ms

108-Bandwidth and delay are:

a) Two unrelated metrics

b) Two performance metrics of a link

c) Always equal

d) Only important for analog signals

109-The Bandwidth-Delay Product defines:

a) The speed of data transfer

b) The amount of data that can be processed in a given time

c) The number of bits that can fill a link

d) The maximum latency of a connection

110- Which of the following communication modes allows data transmission in both directions simultaneously? Select the correct option

A) Full Duplex

B) Simplex

C) Half Duplex

D) Unidirectional

111- Transmission media does not include _____.Select the correct option

a) Air

b) Water

c)None of the above

112-Free s How are links established in computer networking? Select the correct option

a) By creating online profiles on social media platforms.

b) Using various mediums such as wired (e.g., Ethernet cables, fiber-optic cables) or wireless (e.g., Wi-Fi, Bluetooth) technologies.

c) Through software applications and programs.

d) By connecting multiple monitors to a computer setup.pace

113- Which communication mode allows data transmission in one direction only, without any feedback or response from the receiving device? Select the correct option

a) Full Duplex

b) Bidirectional

c) Simplex

d) Half Duplex

114- What does a connecting device do in a network? Select the correct option:

a) Controls network traffic and facilitates data transfer between devices.

b) Determines the type of data transmitted in the network.

c) Provides power supply to network devices.

d) Manages network security.

115- What is an example of a multipoint connection in a network? Select the correct option

a) Multiple devices connecting to the same Wi-Fi router for internet access.

b) A computer connected to a printer via a USB cable.

c) A direct Ethernet cable connection between two computers.

d) Two computers communicating through a dedicated fiber-optic link.

116- Digital transmission involves the transmission of:

A. Analog signals

B. Digital signals

C. Both A and B

D. Neither A nor B

117- Digital to digital conversion is necessary when:

A. The sender and receiver use different digital encoding schemes

- B. The sender and receiver use the same digital encoding scheme
- C. The sender uses an analog signal and the receiver uses a digital signal
- D. The sender uses a digital signal and the receiver uses an analog signal

118-Line coding, block coding, and scrambling are techniques used for:

- A. Analog to digital conversion
- B. Digital to analog conversion

C. Digital to digital conversion

- D. None of the above

119-A data element is the smallest entity that can represent a piece of information.

- A. data element
- B. signal element

C. bit

- D. byte

120-A signal element is the shortest unit of a digital signal.

- A. data element
- B. bit
- C. byte

D. signal element

121-The relationship between data elements and signal elements is:

- A. Data elements are carriers, and signal elements are carried

B. Signal elements are carriers, and data elements are carried

- C. Data elements and signal elements are the same
- D. There is no relationship between data elements and signal elements

122-Line coding is primarily used to:

- A. Convert analog signals to digital signals
- B. Convert digital signals to analog signals

C. Map digital data onto a physical transmission medium

- D. None of the above

123-Block coding is primarily used to:

A. Add redundancy to digital data for error detection and correction

- B. Convert analog signals to digital signals
- C. Map digital data onto a physical transmission medium
- D. None of the above

124-Scrambling is primarily used to:

A. Randomize the pattern of digital data to make it more difficult to intercept

- B. Convert analog signals to digital signals
- C. Map digital data onto a physical transmission medium
- D. None of the above

125-Digital transmission is generally more reliable and efficient than analog transmission.

A. reliable ... efficient

B. less reliable ... less efficient

C. reliable ... less efficient

D. less reliable ... efficient

126- •A data element represents a single bit of information.

A. byte

B. bit

C. signal

D. symbol

127-A signal element is the physical representation of a data element.

A. physical representation

B. logical representation

C. abstract representation

D. symbolic representation

128-The modulation rate (r) is the ratio of data elements to signal elements.

A. modulation rate

B. data rate

C. bit rate

D. baud rate

129-If one data element is represented by one signal element, the modulation rate is:

A. $r = 2$

B. $r = 3$

C. $r = 4$

D. $r = 1$

130-If four data elements are represented by three signal elements, the modulation rate is:

A. $r = 1$

B. $r = 2$

C. $r = 4/3$

D. $r = 3$

131-A higher modulation rate means that more data elements can be represented by a given number of signal elements.

A. more

B. fewer

C. the same number of

D. no relationship between

132-A higher modulation rate can improve the efficiency of data transmission.

A. higher ... efficiency

B. lower ... efficiency

C. higher ... inefficiency

D. lower ... inefficiency

132- Data Rate is defined as:

a) The number of signal elements sent in 1 second

b) The number of data elements sent in 1 second

- c) The number of pulses per second
- d) The number of modulation elements in a signal

133-Signal Rate is also referred to as:

- a) Bit rate
- b) Data rate

c) Baud rate or pulse rate

- d) Frame rate

134-What does line coding do?

- a) Converts analog signals to digital signals
- b) Compresses data

c) Converts a sequence of bits into a digital signal

- d) Measures the frequency of a signal

135-Block coding changes:

- a) A block of n bits into a block of m bits ($n < m$)
- b) A block of m bits into a block of n bits ($n < m$)

c) A block of m bits into a block of n bits ($n > m$)

- d) A block of m bits into n blocks ($n > m$)

136-Why do we need redundancy in block coding?

a) To ensure synchronization

- b) To increase the bandwidth
- c) To reduce noise
- d) To compress data

137-One advantage of block coding is that it:

- a) Reduces data rate
- c) Increases latency
- c) Reduces signal rate

d) Improves line coding performance

138-Data rate is also known as:

- a) Baud rate

b) Bit rate

- c) Pulse rate
- d) Frame rate

139-What is the purpose of the mB/nB encoding technique?

- a) To reduce signal noise

b) To ensure synchronization through redundancy

- c) To decrease signal rate
- d) To improve data compression

140-Digital data in a computer is stored as:

- a) Blocks of analog signals

b) Modulated waves

c) Sequences of bits

d) Sequences of symbols

141-Block coding creates redundancy by:

a) Changing n bits to m bits ($m > n$)

b) Changing m bits to n bits ($n > m$)

c) Removing synchronization bits

d) Compressing data blocks

142-NRZ-I combined with block coding solves synchronization issues, but it has:

a) Bandwidth issues

b) Long series of 0s

c) A DC component

d) No redundancy

143-Bipolar AMI has which of the following characteristics?

a) Wide bandwidth and no synchronization issues

b) DC component and no synchronization issues

c) Narrow bandwidth and no DC component

d) Narrow bandwidth and no synchronization issues

144-What is the purpose of scrambling techniques like B8ZS and HDB3?

- a) To reduce noise in the signal
- b) To increase data rate
- c) To improve compression

d) To insert pulses based on defined scrambling rules

145-Bipolar with 8-Zero Substitution (B8ZS) is used to:

- a) Remove synchronization issues

b) Insert pulses to replace long sequences of 0s

- c) Eliminate the DC component
- d) Reduce bandwidth

146-High-Density Bipolar 3-Zero (HDB3) is another scrambling technique used to:

a) Replace sequences of zeros with a pattern to maintain synchronization

- b) Reduce the data rate
- c) Increase bandwidth
- d) Insert DC components in the signal

147-Digitization of analog data involves which two techniques?

- a) NRZ-I and HDB3

b) Pulse Code Modulation (PCM) and Delta Modulation (DM)

- c) B8ZS and AMI
- d) NRZ and block coding

148-The Nyquist theorem states that the sampling rate must be:

- a) Equal to the signal frequency

b) Half of the signal frequency

c) Twice the highest frequency of the signal ($2f_h$)

d) Four times the signal frequency

149-If the sampling rate is 2 times the Nyquist rate, it is:

a) $f_s = 4f$

b) $f_s = 2f$

c) $f_s = f$

d) $f_s = 0.5f$

150-Which of the following is NOT true about the Nyquist theorem?

a) $f_s = 2f_h$ is the minimum sampling rate for accurate digitization

b) $f_s = f$ (one-half the Nyquist rate) ensures accurate digitization

c) Sampling at $f_s = 4f$ provides more data points for a sine wave

d) $f_s = 2f$ is the Nyquist rate

151- Pulse Code Modulation (PCM) involves which of the following processes?

a) Sampling and decoding

b) Sampling, quantization, and encoding

c) Quantization and transmission

d) Encoding and modulation

152-In PCM, the sampling output is:

a) Encoded immediately

b) Quantized into specific levels based on amplitude

c) Sent as is without any quantization

d) Converted into analog signals

153-Delta Modulation (DM) differs from PCM in that it:

a) Finds the change from the previous sample

b) Finds the exact value of each sample

c) Is more complex than PCM

d) Uses code words to represent values

154-In parallel transmission, data is sent:

a) One bit at a time

b) Multiple bits at a time using multiple wires

c) Over a single communication channel

d) Using code words

155-In serial transmission, data is sent:

a) One bit at a time over a single communication channel

b) Multiple bits at a time using multiple wires

c) Only for audio and video signals

d) With real-time synchronization

156-Isochronous transmission is primarily used for:

a) Data synchronization

b) Non-real-time data transfer

c) Real-time audio and video data transmission

d) Parallel transmission

157-Which of the following is true for PCM?

a) It is simpler than DM

b) It is used for encoding changes between samples

c) It finds the value of the signal amplitude for each sample

d) It transmits one bit at a time

158-In Delta Modulation (DM), what is not used?

a) Sample values

b) Code words

c) Amplitude levels

d) Sampling

159-In parallel transmission, how many wires are required to send 'n' bits at a time?

a) 1 wire

b) $n/2$ wires

c) n wires

d) No wires required

160-Serial transmission is preferred over parallel transmission for:

a) Long-distance communication

- b) High-speed local communication
- c) Multi-channel data transfer
- d) Sending multiple bits at a time

161- Which characteristic of a sine wave is altered in Amplitude Shift Keying (ASK)?

- a) Frequency
- b) Phase
- c) Bandwidth

d) Amplitude

162-What happens to frequency and phase during Amplitude Shift Keying (ASK)?

- a) Frequency changes, phase remains constant

b) Both remain constant

- c) Both change
- d) Frequency remains constant, phase changes

163-Which of the following modulation techniques alters the phase of the signal to represent data?

- a) ASK

b) PSK

- c) FSK
- d) OOK

164-In digital-to-analog conversion, which factor defines the rate at which signal elements are transmitted?

- a) Bit rate

b) Carrier signal

c) Baud rate

d) Bandwidth

165-Binary ASK is also known as:

a) Frequency Modulation

b) On-Off Keying (OOK)

c) Quadrature Amplitude Modulation

d) Phase Modulation

165-Which combination of modulation techniques could be used to increase the bit rate of a signal?

a) ASK and FSK

b) FSK and PSK

c) ASK and PSK

d) Only PSK

166-The baud rate in analog transmission of digital data is typically:

a) Higher than the bit rate

b) Less than or equal to the bit rate

c) Always equal to the bandwidth

d) Higher than the carrier signal

167-What is the role of the carrier signal in digital-to-analog conversion?

a) It is modulated by changing its amplitude, frequency, or phase

b) It changes the data elements

- c) It increases the bit rate
- d) It is independent of the data signal

168-Which modulation technique is affected by the equipment's ability to distinguish small differences in phase?

a) ASK

b) PSK

c) FSK

d) OOK

169-Simultaneous transmission of multiple signals across a single data link is used to:

a) Increase bandwidth utilization

b) Decrease the baud rate

c) Ensure constant amplitude

d) Use a single carrier signal for each channel

170- Multiplexing is a technique used to:

A. Increase the speed of a network

B. Transmit multiple signals over a single channel

C. Reduce the bandwidth of a channel

D. None of the above

171-Frequency-division multiplexing (FDM) is primarily used for:

A. Digital signals

B. Analog signals

- C. Both analog and digital signals
- D. Neither analog nor digital signals

172-In FDM, each sending device modulates a different:

- A. Amplitude
- B. Phase

C. Carrier frequency

- D. None of the above

173-Wavelength-division multiplexing (WDM) is primarily used for:

A. Optical fiber communications

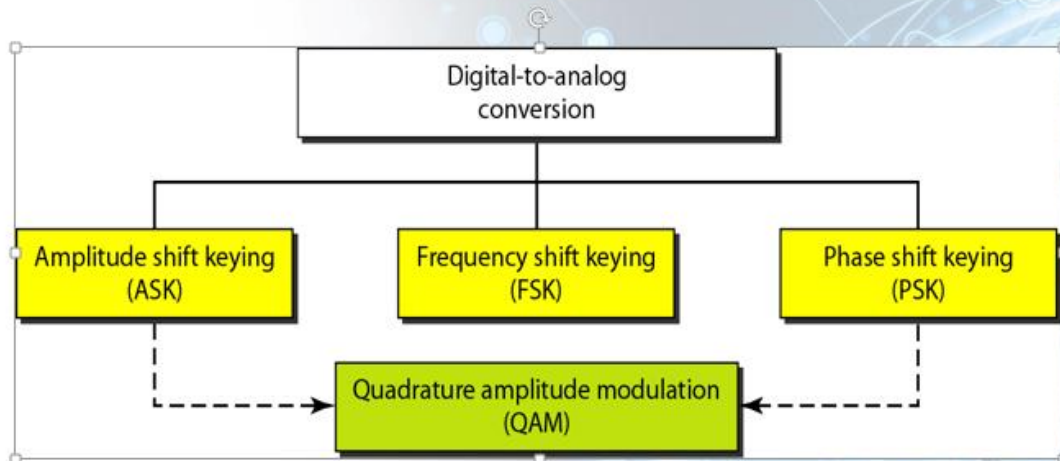
- B. Copper wire communications
- C. Wireless communications
- D. None of the above

174-Time-division multiplexing (TDM) is primarily used for:

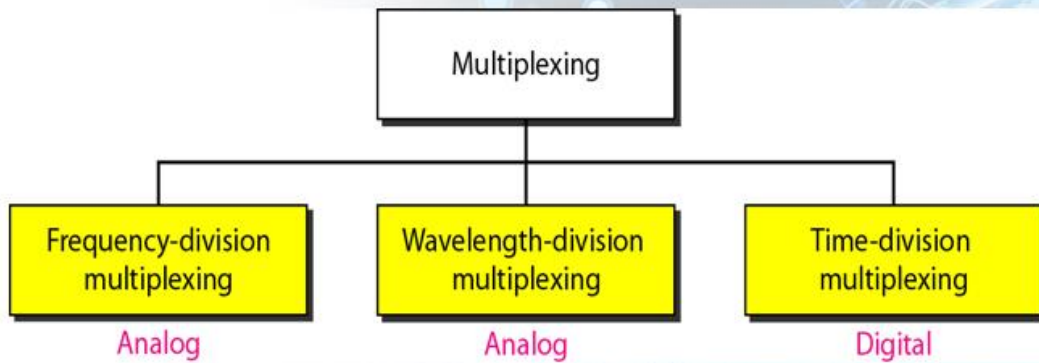
A. Digital signals

- B. Analog signals
- C. Both analog and digital signals
- D. Neither analog nor digital signals

Types of Digital to Analog Conversion



Categories of Multiplexing



175- Frequency-division multiplexing (FDM) is a technique used to:

A. Combine multiple analog signals into a single composite signal

B. Separate multiple analog signals from a single composite signal

C. Convert analog signals to digital signals

D. None of the above

174-In FDM, each sending device modulates a different:

A. Amplitude

B. Phase

C. Carrier frequency

D. None of the above

176-The multiplexed signals in FDM are combined into a single composite signal that can be transported by the link.

A. combined

B. separated

C. amplified

D. attenuated

177-FDM is primarily used for:

A. Digital signals

B. Analog signals

C. Both analog and digital signals

D. Neither analog nor digital signals

178-A disadvantage of FDM is that it can be inefficient if the bandwidth of the link is much larger than the combined bandwidths of the signals.

A. inefficient

B. efficient

C. effective

D. ineffective

179- In FDM, the minimum bandwidth required for a link must be greater than the combined bandwidths of the signals to be transmitted.

A. less than

B. greater than

C. equal to

D. none of the above

180-Guard bands are used in FDM to prevent interference between the multiplexed signals.

A. prevent interference

B. reduce noise

C. increase bandwidth

D. none of the above

181-The bandwidth of a link in FDM should be calculated by adding the bandwidths of all the signals to be transmitted, plus the guard bands.

A. calculated

B. estimated

C. measured

D. none of the above

182-Wavelength Division Multiplexing (WDM) is designed to:

- a) Reduce the data rate of fiber-optic cables
- b) Increase the data rate of metallic cables

c) Use the high-data-rate capability of fiber-optic cables

- d) Decrease the bandwidth of fiber-optic cables

183-Why is using a fiber-optic cable for a single line considered inefficient?

a) It wastes the available bandwidth

- b) Fiber-optic cables have low bandwidth
- c) Fiber-optic cables cannot support high data rates
- d) Multiplexing is not supported by fiber-optic cables

184-Time Division Multiplexing (TDM) works by:

a) Allowing multiple connections to share the bandwidth by allocating time slots

- b) Dividing the frequency of the link among multiple connections
- c) Using different wavelengths for different connections
- d) Using the same time slot for all connections

185-TDM is best described as a:

a) Digital process that shares the bandwidth of a link by dividing time

- b) Technique that shares the link based on different wavelengths
- c) Method to reduce the data rate of connections
- d) Process to reduce the bandwidth of the link

186-In spread spectrum techniques, which is one of the main goals?

- a) Reduce the redundancy in the signal
- b) Ensure constant transmission speed

c) Prevent interception and jamming

- d) Share bandwidth equally among stations

187-Which spread spectrum technique involves modulating different carrier frequencies at different times?

a) Frequency Hopping Spread Spectrum (FHSS)

- b) Direct Sequence Spread Spectrum (DSSS)
- c) Wavelength Division Multiplexing (WDM)
- d) Time Division Multiplexing (TDM)

188-What does Frequency Hopping Spread Spectrum (FHSS) do with the carrier frequency?

- a) It remains constant throughout the transmission

b) It hops between different frequencies at different times

- c) It changes based on the data rate
- d) It is only used in fiber-optic transmission

189-Direct Sequence Spread Spectrum (DSSS) achieves security by:

- a) Modulating the signal using different wavelengths
- b) Allocating specific time slots to each connection

c) Spreading the signal over a wide frequency band

- d) Using low bandwidth to reduce interference

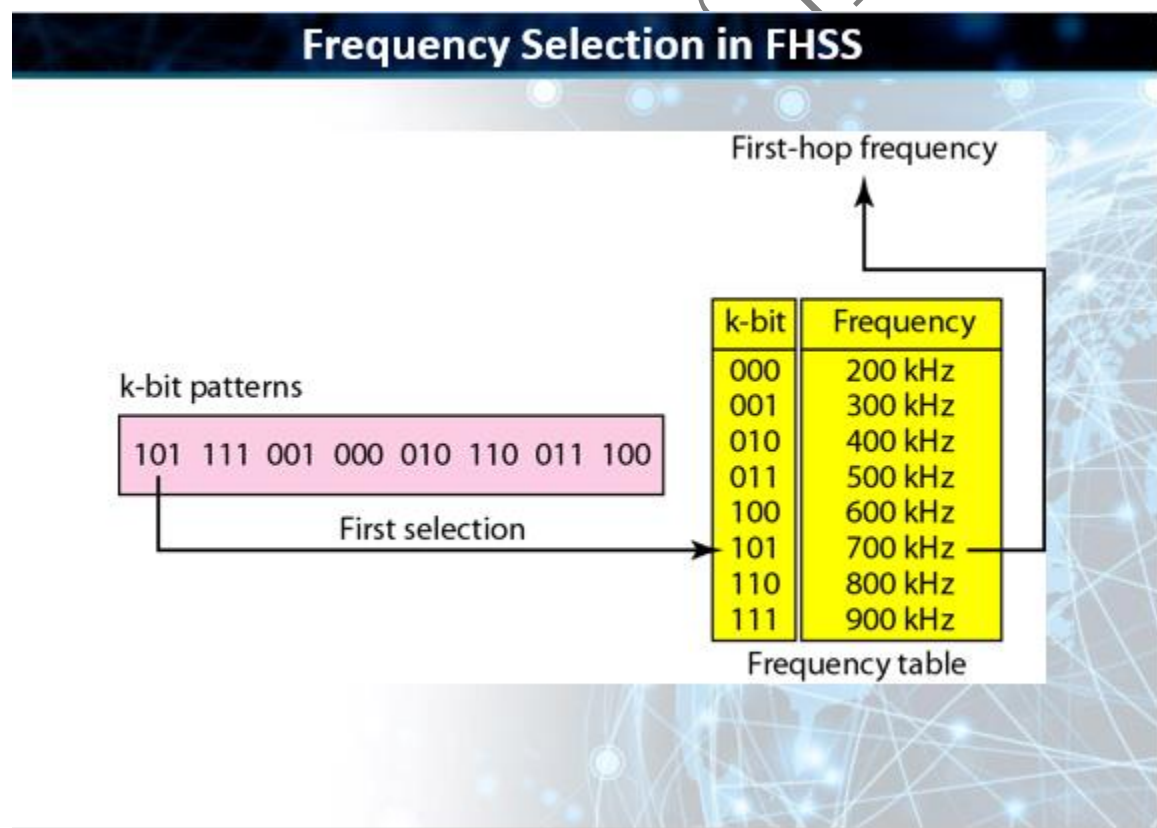
190-Which of the following is true about spread spectrum techniques?

a) They require a larger bandwidth than what is typically needed

- b) They are used to reduce the data rate
- c) They can only be used in wired networks
- d) They are based on time-sharing among connections

191-In wireless communications, spread spectrum techniques help to:

- a) Increase the data rate
- b) Prevent eavesdropping and jamming**
- c) Allow more connections to share the same bandwidth
- d) Minimize latency during transmission



192- In Direct Sequence Spread Spectrum (DSSS), what is used to replace each data bit?

- a) Frequency Hopping
- b) Bandwidth Expansion

c) 'n' bits using a spreading code

- d) Time Division Multiplexing

193-In DSSS, each bit is replaced by 'n' bits. These 'n' bits are referred to as:

- a) Packets

b) Chips

- c) Waves
- d) Bytes

194-Transmission media are located:

- a) At the application layer
- b) At the transport layer

c) Below the physical layer (Layer 0)

- d) At the network layer

195-Which of the following is a guided transmission medium?

- a) Air
- b) Vacuum

c) Twisted-pair cable

- d) Free space

196-Twisted-pair cables consist of:

a) Two copper conductors twisted together, each with plastic insulation

b) A single copper conductor wrapped in plastic

c) A glass or plastic core transmitting light

d) Coaxial wires arranged in parallel

197-The main function of the second wire in a twisted-pair cable is to:

a) Transmit data

b) Act as a power supply

c) Serve as a ground reference

d) Filter interference

198-Coaxial cables are generally used for:

a) Low-frequency signals

b) Higher frequency signals compared to twisted-pair cables

c) Light signal transmission

d) Power distribution

199-Fiber-optic cables transmit signals using:

a) Electrical currents

b) Magnetic waves

c) Light signals through glass or plastic

d) Radio waves

200-What happens to light when it moves from one medium to another with a different density in a fiber-optic cable?

a) The light ray changes direction

b) The light ray stops traveling

c) The light ray speeds up

d) The light ray stays in a straight line

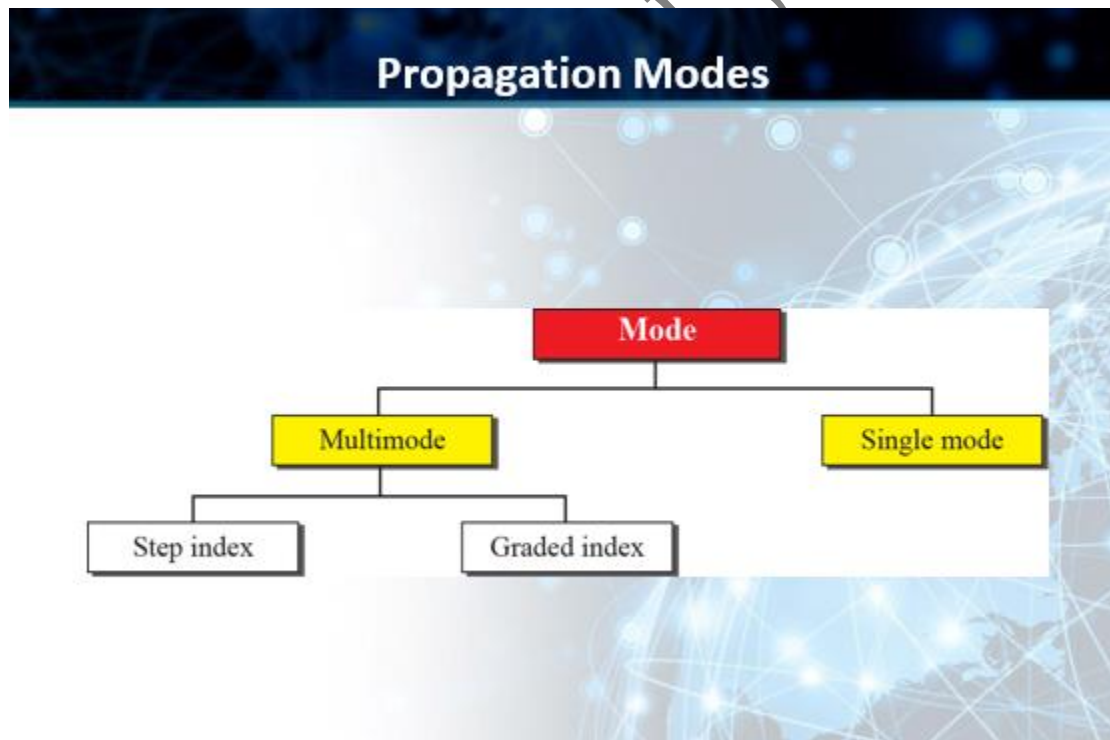
201-Which of the following media is NOT considered a guided transmission medium?

a) Fiber-optic cable

b) Twisted-pair cable

c) Air

d) Coaxial cable



202- Fiber optic cables are made of:

A. Copper

B. Glass or plastic

C. Aluminum

D. None of the above

203-Fiber optic cables transmit signals in the form of:

A. Electricity

B. Light

C. Radio waves

D. None of the above

204-A major advantage of fiber optic cables is their:

A. High bandwidth

B. Low bandwidth

C. High attenuation

D. Susceptibility to electromagnetic interference

205-Fiber optic cables have less attenuation compared to copper cables.

A. less

B. more

C. the same

D. none of the above

206-A disadvantage of fiber optic cables is their:

A. High installation cost

B. Low bandwidth

C. High attenuation

D. Low weight

207-Unguided media transport waves without using a:

A. Physical conductor

B. Cable

C. Wire

D. All of the above

208-Unguided media is often referred to as:

A. Wired communication

B. Wireless communication

C. Optical communication

D. None of the above

209-Signals transmitted through unguided media are typically:

A. Confined to a specific path

B. Broadcast through free space

C. Limited to a small area

D. None of the above

210-Radio waves are electromagnetic waves with frequencies between:

A. 100 kHz and 1 GHz

B. 3 kHz and 1 GHz

C. 1 GHz and 300 GHz

D. 300 GHz and 1 THz

211- Microwaves are electromagnetic waves with frequencies between:

A. 100 kHz and 1 GHz

B. 3 kHz and 1 GHz

C. 1 GHz and 300 GHz

D. 300 GHz and 1 THz

212-Microwaves are:

A. Omnidirectional

B. Unidirectional

C. Both omnidirectional and unidirectional

D. Neither omnidirectional nor unidirectional

213-Microwaves can be:

A. Narrowly focused

B. Widely dispersed

C. Both narrowly focused and widely dispersed

D. Neither narrowly focused nor widely dispersed

214-Infrared Waves: Infrared waves have frequencies between:

- A. 100 kHz and 1 GHz
- B. 3 kHz and 1 GHz
- C. 1 GHz and 300 GHz

D. 300 GHz and 400 THz

215-Infrared waves can be used for:

- A. Long-range communication

B. Short-range communication

- C. Both long-range and short-range communication
- D. Neither long-range nor short-range communication

216-Infrared waves cannot penetrate:

A. Walls

- B. Windows
- C. Doors

All of the above

217-A disadvantage of infrared waves is their:

A. Limited range

- B. High frequency
- C. Low frequency
- D. None of the above

218-Infrared waves can be used to prevent interference between different systems.

A. prevent interference

B. cause interference

C. have no effect on interference

D. None of the above

219-Transmission medium are of _____ types.Choose the correct answer:

(A)4

(B)2

(C)3

(D)1

220-_____ means loss of energy in signal. Select the correct option

(a)Attenuation

(b)Distortion

(c)Noise

(d)Delay

221- In transmission impairments _____ noise occurs from other sources like motors and appliances etc.Choose the correct answer:

(A)Thermal

(B)Impulse

(C)Cross talk

(D)Induced

222-Taking off the header from the message is called _____.Choose the correct answer:

(A)Decapsulation

(B)Encryption

(C)Stuffing

(D)Layering

223-Two PCs and one printer are connected in a network within a room; it is an example of _____.

Choose the correct answer:

(A)SONET

(B)WAN

(C)MAN

(D)LAN

224-_____ is the Nyquist bit rate formula for noiseless channel.

Choose the correct answer:

(A) $\text{BitRate} = 2 * \text{Bandwidth} * L$

(B) $\text{BitRate} = 2 * \text{Bandwidth} * \log_2 L$

(C) $\text{BitRate} = 2 * \text{Bandwidth} * \log_2$

(D) $\text{BitRate} = 3 * \text{Bandwidth} * \log_2 L$

225-Set of rules to be followed for effective network communication is called _____.

Choose the correct answer:

- (A)Topology
- (B)Encryption
- (C)Protocol**
- (D)Switching

226-Bandwidth can be measured in _____.Choose the correct answer:

- (A)Phase
- (B)Decibel
- (C)Bps**

227-In analog transmission of digital data, the required bandwidth is always proportional to the signal rate except in _____, a digital to analog conversion technique.

Choose the right option:

- (A)FSK**
- (B)ASK
- (C)PSK
- (D)NSK

228-In scrambling coding scheme, the number of pulses replacing the bits will be _____ ? Choose the right option:

- (A)Half of the pulses
- (B)Equal to pulses**
- (C)Double the pulses
- (D)Triple the pulses

229- When the bandwidth of a link is greater than the combined bandwidths of the signals, we use _____ multiplexing. Choose the right option:

- (A)Time division
- (B)Hybrid division
- (C)Wavelength division

(D)Frequency division

230-Time division multiplexing is used in _____ systems .Choose the right option:

- (A)Analog
- (B)Automated

(C)Digital

- (D)Hybrid

231-The conversion of analog sample of the signal into digital form is called _____ process. Choose the right option:

- (A)Quantizing
- (B)Sampling
- (C)Modulation

(D)Quantizing and Sampling

232-In _____ encoding scheme, the voltage level oscillates between a positive and a negative value although it may remain at zero level between the two values. Choose the right option:

- (A)Uni-Polar

(B)Bi-Polar

- (C)Non-Polar
- (D)Polar

233-Unit of Phase is _____.Choose the right option:

(A)Degree

(B)watts

(C)bauds

(D)Bits per second

234-Which of following technique(s) is used for Analog-to-Digital Conversion.Choose the right option:

(A)Bit Modulation(BDM)

(B)Pulse Code Modulation (PCM)

(C)Delta Modulation (DM)

(D)Both Pulse Code and Delta Modulation

235- in Manchester and differential Manchester encoding schemes, the signal rate is _____ than that of NRZ. Choose the right option:

(A)Double

(B)Triple

(C)lower

(D)Equal

IMP NOTE: *"Hope this file will be helpful to you all. If you are looking for more books, mid-term and final-term lecture-wise files, then join our VU Road to Finals group."*

Link is here :

<https://chat.whatsapp.com/CU4hgl67nW0FcxZpKDSQhV>

Regards by Harry

VU ROAD TO FINALS HARRY FILE